

Dundee Zhang

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EDUCATION

University of Waterloo

Bachelor's of Computer Science

Waterloo, ON

Sep 2024 – Apr 2028

- **Societies:** The Mathematics Society - **Councilor** and **Director**.
- **Clubs:** Computer Science Club - **Web Developer**, **Office Manager**, and **President**.
- **Design Teams:** Electrium Mobility - **Web Developer** - Wrote scripts to allow renting vehicles on the store.

EXPERIENCE

airfairness | *Software Developer*

May 2025 – Present

- Designed the **Human Interaction** framework powering key product workflows, optimizing for **modularity** and **speed**.
- Engineered full-stack features using **Next.js** and **Convex** within a **monorepo**, streamlining development across services.
- Collaborated via **GitHub** to **review code** and ensure **smooth CI/CD** in a fast-moving startup environment.

University of Waterloo Computer Science Club | *President*

May 2025 – Present

- Lead a team of **60+ students**, overseeing club operations, strategic planning, and campus-wide initiatives.
- Manage **human resources**, support executive teams, and ensure smooth coordination across all subcommittees.
- Assist with **technical**, **logistical**, and **outreach** efforts across web, office, marketing, etc teams to keep operations aligned.

Ignition Hacks | *Mentor*

Aug 2024

- Mentored over **500** students in **JavaScript**, **TypeScript** and **Python** with **React** and **Flask**.
- Facilitated one-on-one sessions, empowering students with **hands-on** experience with **AI**, **CV**, and **NLP**.
- Assisted to integrate **Language Models** and **Computer Vision** into projects with endpoint (**REST**) APIs.

NRGHacks | *Co-Founder and Judge*

Aug 2023 – Jun 2024

- Established a **school-board** and **SHSM recognized** hackathon for students within the York Region.
- Collaborated with **educators** and **professionals** to create a learning and networking environment for participants.
- Raised over **\$1000** for multiple high school computer science clubs totaling over **200 hackers** overall.

PROJECTS

EcoElo | *React, JavaScript, PostgreSQL, Resend, Regex*

- Designed front-end structure and **routing** for pages, boosting **development efficiency** by over **300%**.
- Utilized Resend to implement **magic-links** to streamline user login and sign-up speeds by **100%**.
- Integrated a **SQL database** to create a point system to allow user-rankings (elo system) and rewards.

Professional Career Assistant | *Python, JavaScript, Flask, Firebase, HTML, CSS, OpenAI API*

- Achieved over **20** unique user **sign-ups** within the first **24 hours** of product launch.
- Integrated **AI** into a professional career development toolkit using **Flask** and **JavaScript** in **backend**.
- Leveraged **Firebase** to ensure the **security** of the app and to **streamline** user sign-up experience by **400%**.

AI-Enhanced To Do List | *Python, Flask, SQLAlchemy, OpenAI API, HTML, CSS*

- Designed an **AI powered** To-Do list that **automatically generates** subtasks for the users tasks.
- Engineered prompts to achieve **100% realistic** and **useful** subtasks for the user to complete.
- Incorporated **SQLAlchemy** database to store the task list locally to ensure usefulness and **privacy**.

GoTerpret Bash Shell | *Go*

- Built a **Bash shell interpreter** from scratch using **Go**, allowing shell **commands**, **piping**, and **redirection**.
- Implemented built-in commands such as **cd**, **exit**, and **echo**, providing a simplified shell environment for users.
- Integrated **error handling** to ensure proper execution and **improve** the **stability** of the shell interpreter.

Portfolio and Blog | *TypeScript, Node, React, CSS, HTML, Markdown*

- Constructed portfolio and blog with **user commenting**, **contact form**, and **markdown interpreter**.
- Crafted **reactive** and playful **animated** elements throughout the webpages with **mobile compatibility**.
- Automated **CI/CD** through **Cloudflare Workers and Pages** to allow for easy website updates/tweaks.

AWARDS

Rich Media - Aivars Petersons Memorial Award | *University of Waterloo*

Jun 2024

- **One award**, given annually to an **outstanding** undergraduate student entering Year One of any program in the David R. Cheriton School of Computer Science.

SKILLS & INTERESTS

Languages: C, C++, Go, HTML/CSS, Java, JavaScript, Objective-C, Python, Racket, Scheme, Swift, TypeScript

Frameworks: Bootstrap, Flask, Java Swing, MongoDB, Next.js, React, SQLAlchemy, TailwindCSS

Tools: Bash/Linux Shell, Cloudflare, Docker, Figma, Firebase, Git, Node.js, Vercel, Vim, Vite, VS Code, XCode